

Full Stack Development with MERN Project

Documentation format

1. Introduction

- **Project Title:** SHOPEZ – E-Commerce Application
- **Team Members:** Manideepsai Ragini (Developer, Designer, Tester)

2. Project Overview

- **Purpose:**

SHOPEZ is a full-stack e-commerce web application developed using the MERN Stack. The platform enables users to browse products, search and filter items, add products to cart, place orders, and manage their profiles. Administrators can manage products through a dedicated admin dashboard.

- **Features:**

- User Registration and Login
- JWT Authentication
- Product Listing
- Product Search
- Category Filtering
- Product Details Page
- Shopping Cart
- Order Placement
- Order History
- User Profile Dashboard
- Admin Product Management
- Responsive User Interface

3. Architecture

- **Frontend:** The frontend is developed using React.js and React Router. It provides an interactive user interface for browsing products, managing carts, viewing orders, and accessing user profiles.
- **Backend:** The backend exposes REST APIs for authentication, products, carts, and orders. Express.js handles routing and business logic.
- **Database:** MongoDB stores application data in collections:
 - Users
 - Products
 - Carts
 - Orders

4. Setup Instructions

- **Prerequisites:**
 - Node.js
 - npm
 - MongoDB
 - Git
- **Installation:**
 - **Clone Repository**
 - `git clone <repository-url>`
 - **Backend Setup**
 - `cd server`
 - `npm install`
 - Create a `.env` file:
 - `PORT=5000`
 - `MONGO_URI=your_mongodb_connection_string`
 - `JWT_SECRET=your_secret_key`
 - **Frontend Setup**
 - `cd client`
 - `npm install`

5. Folder Structure

- **Client:** client/
 - ├── src/
 - ├── components/
 - ├── pages/
 - ├── services/
 - ├── assets/
 - ├── App.jsx
 - └── main.jsx
- **Server:** server/
 - ├── controllers/
 - ├── middleware/
 - ├── models/
 - └── routes/

```
|— config/
|— server.js
└— .env
```

6. Running the Application

- **Frontend:** cd server
npm start
Server runs on:
http://localhost:5000
.
- **Backend:** cd client
npm run dev
Application runs on:
http://localhost:5173

7. API Documentation

Authentication APIs

- **Register User**
- POST /api/auth/register
- Request:
 - {
 - "name": "Ragini",
 - "email": "ragini@gmail.com",
 - "password": "123456"
 - }
- **Login User**
- POST /api/auth/login

Product APIs

- **Get All Products**
- GET /api/products
- **Get Single Product**
- GET /api/products/:id
- **Add Product (Admin)**
- POST /api/products
- **Update Product**
- PUT /api/products/:id
- **Delete Product**
- DELETE /api/products/:id

Cart APIs

- **Add To Cart**
- POST /api/cart
- **Get Cart**
- GET /api/cart
- **Remove Cart Item**
- DELETE /api/cart/:id

Order APIs

- **Place Order**
- POST /api/orders
- **Get Orders**
- GET /api/orders

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8. Authentication

Authentication is implemented using JSON Web Tokens (JWT).

Process

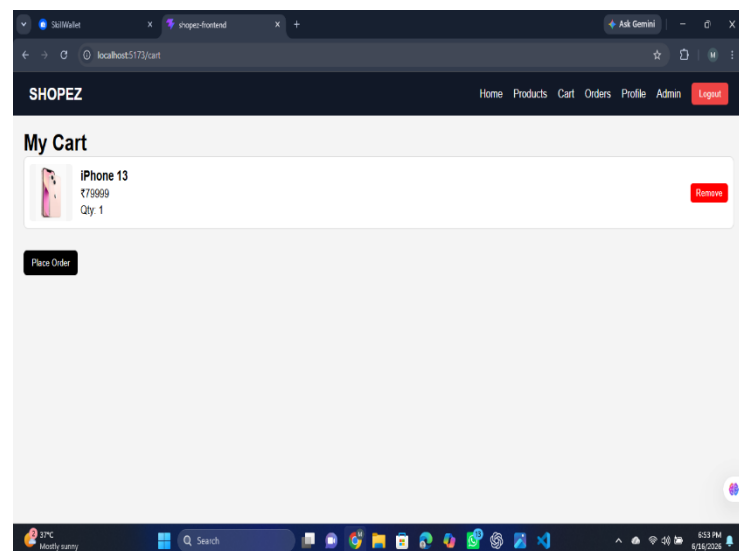
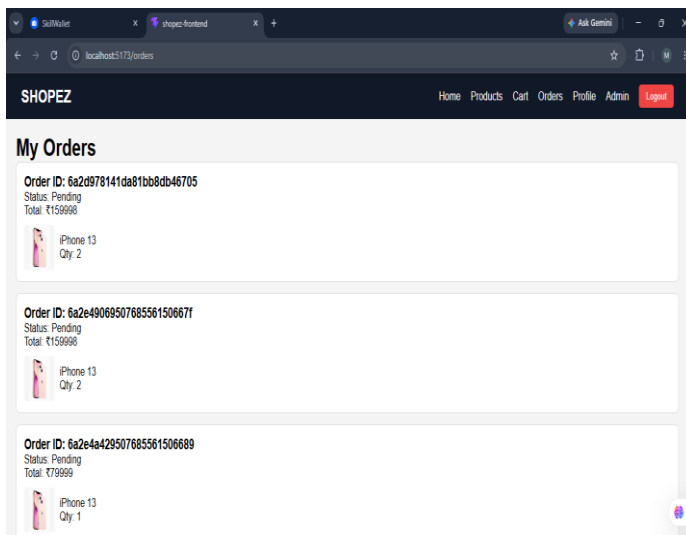
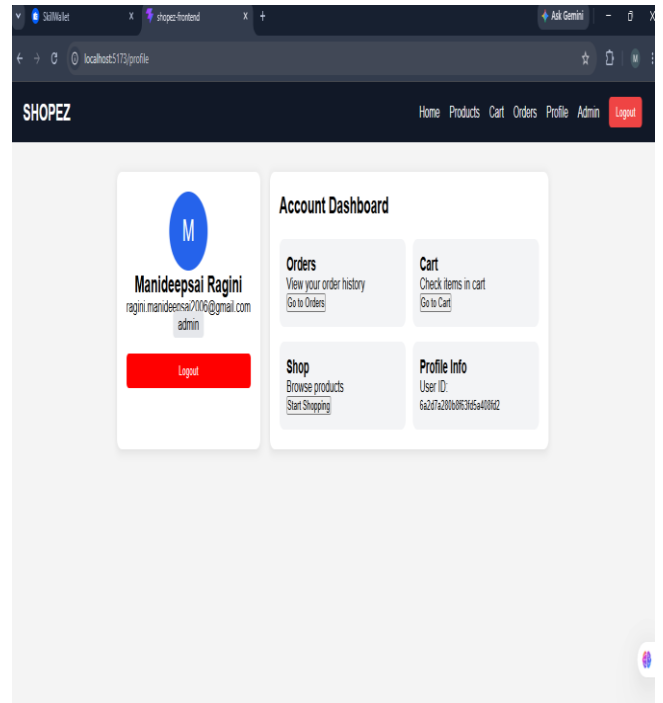
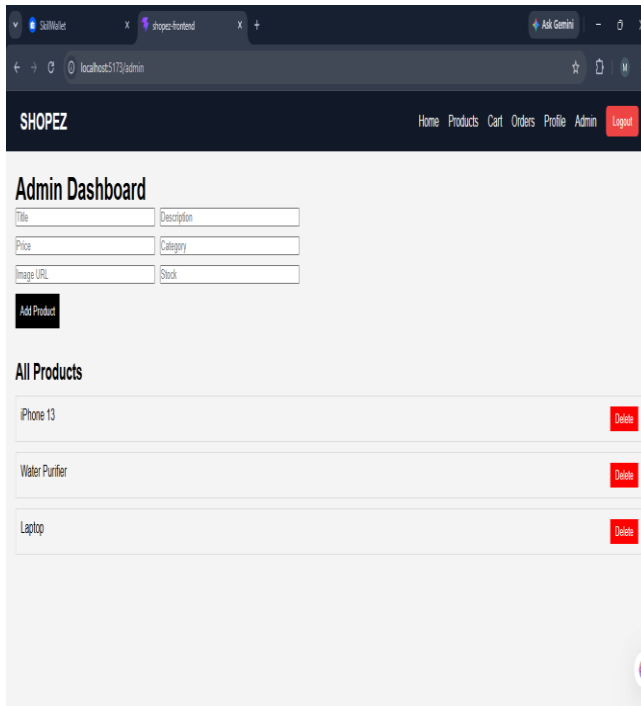
1. User registers or logs in.
2. Server generates JWT token.
3. Token is stored in localStorage.
4. Protected routes validate JWT token.
5. Authorized users can access secured resources.

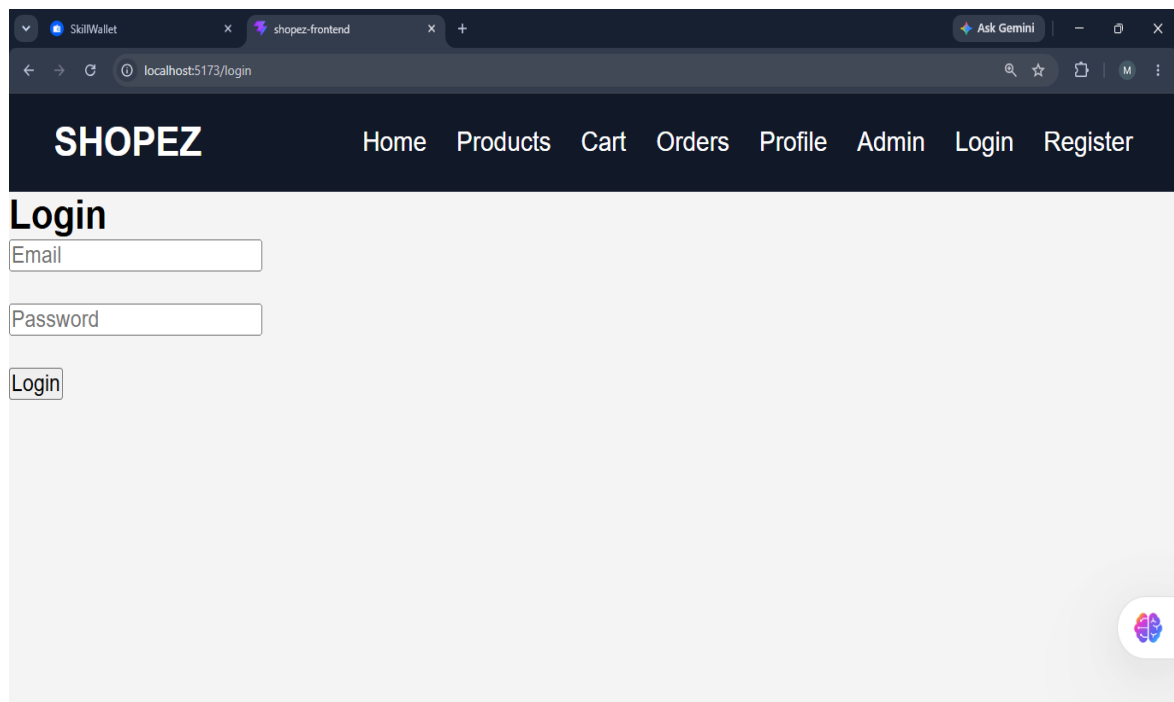
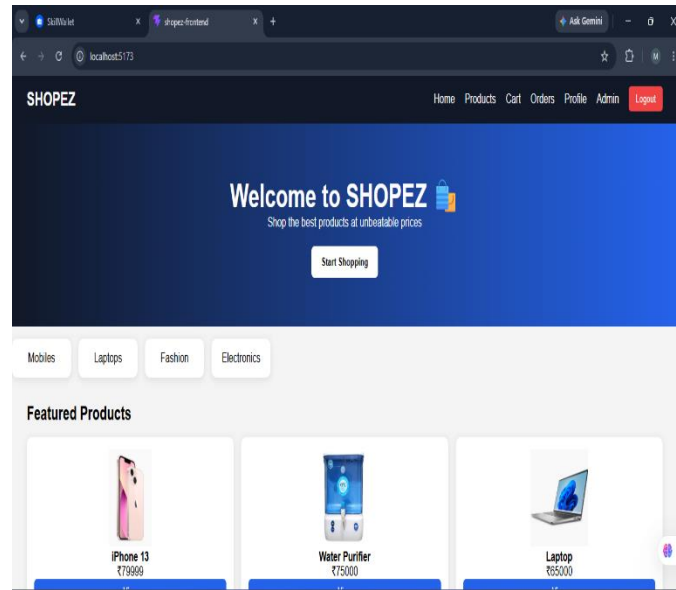
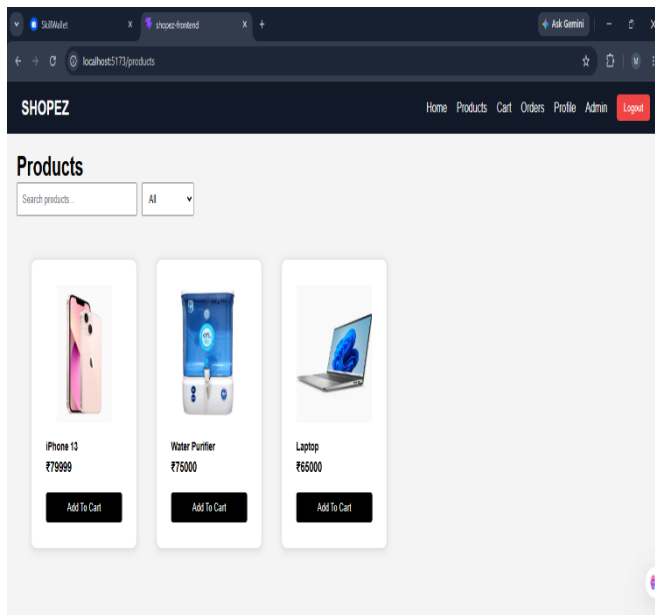
Authorization

- User Role
- Admin Role

Admins can create, update, and delete products.

9. User Interface





10. Testing

Tools Used

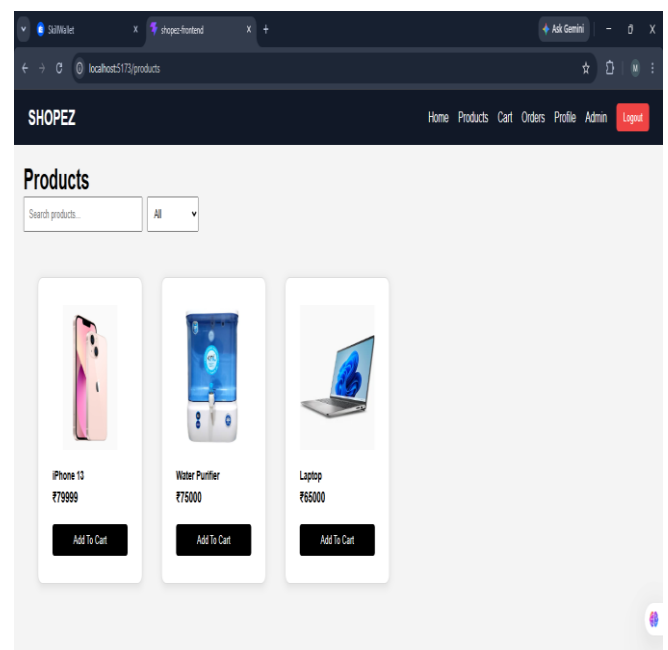
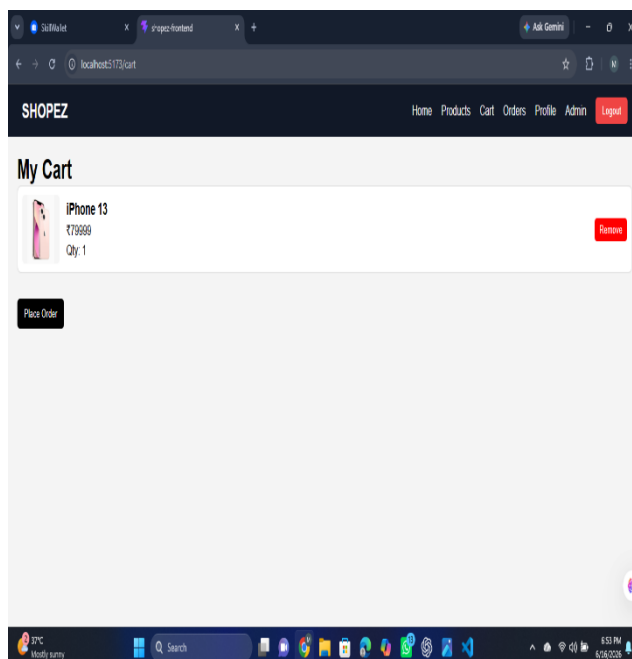
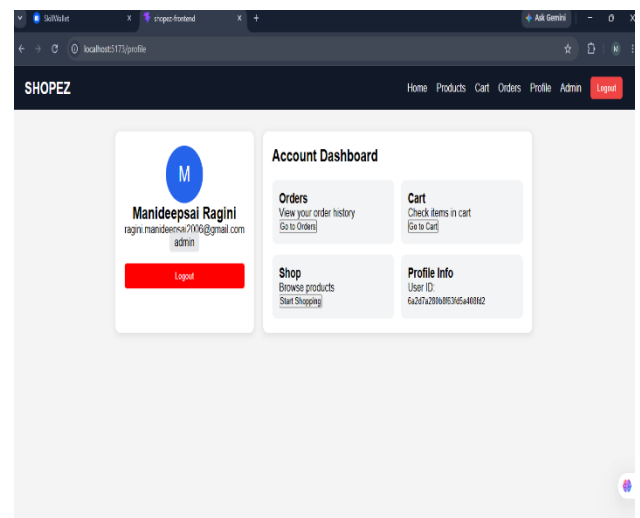
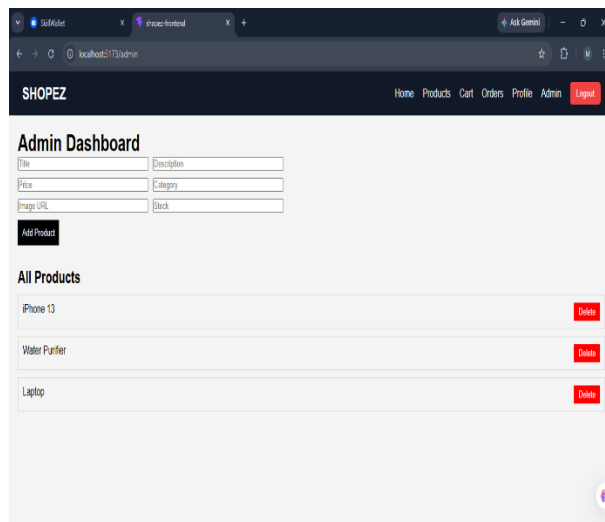
- Thunder Client
- Browser Developer Tools

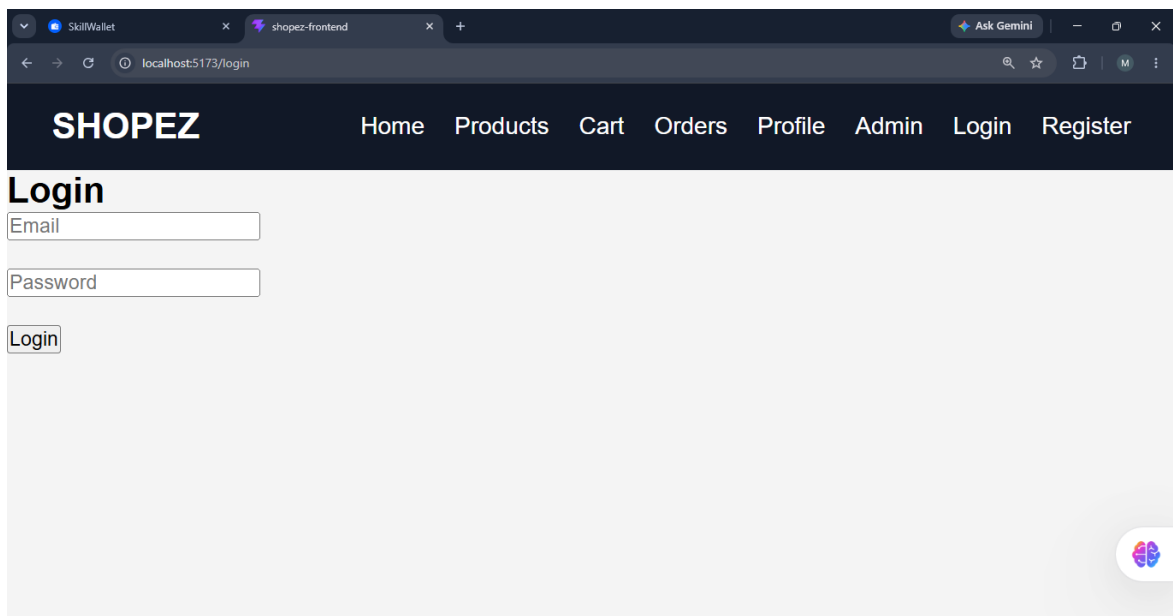
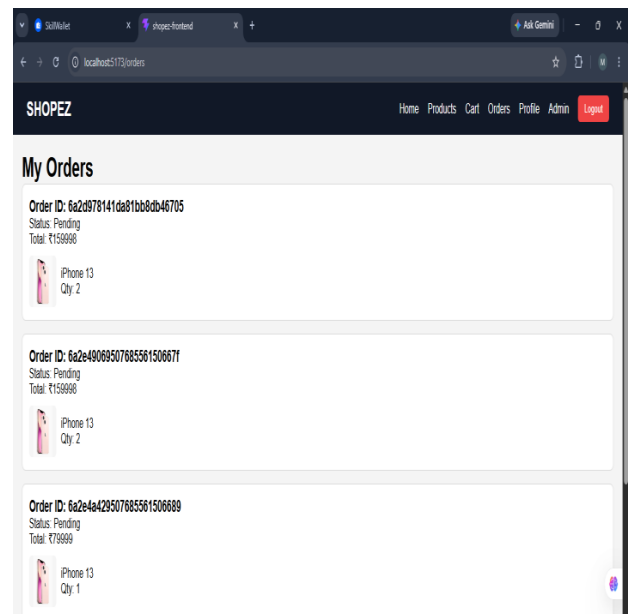
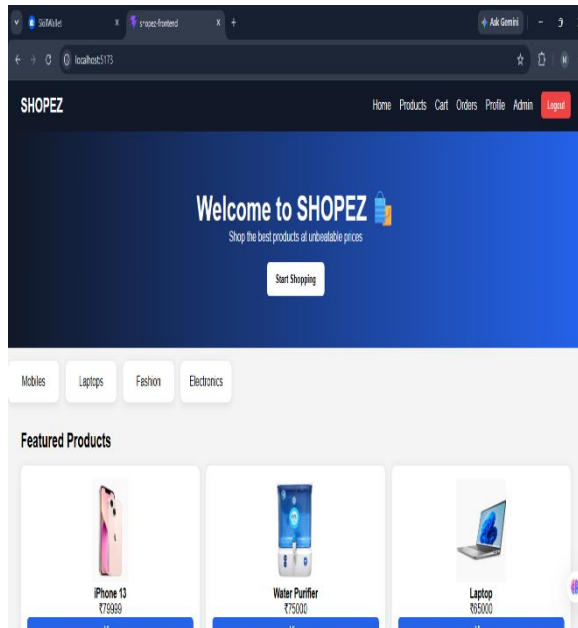
Testing Performed

- Authentication Testing

- Product API Testing
- Cart Testing
- Order Testing
- UI Testing
- All critical functionalities were tested successfully.

11. Screenshots or Demo





12. Known Issues

- Payment gateway not integrated.
- Product reviews not implemented.
- Wishlist functionality not available.
- Email notifications not implemented.

13. Future Enhancements

- Payment Gateway Integration
- Product Reviews & Ratings
- Wishlist Feature
- Recommendation System
- Cloud Deployment

- Mobile Application
- Multi-Vendor Support
- Email Notifications.